MILITARY SPECIFICATION

MOTOR-GENERATOR, 60-CYCLE, AC TO 400-CYCLE AC

(VOLTAGE AND FREQUENCY REGULATED)

SHIPBOARD SERVICE

This amendment forms a part of Military Specification MIL-M-19633B(SHIPS), 10 March 1967.

Pages 1 and 2

2.1, under "SPECIFICATIONS, MILITARY" add:

"MIL-H-46855 - Human Engineering Requirements for Military Systems Equipment and Facilities."

2.1, add the following Military standards, handbooks and publications:

"STANDARDS

MILITARY
MIL-STD-415 - Test Points and Test Facilities for Electronic Systems and Associated Equipment, Design Standard for.
MIL-STD-470 - Maintainability Program Requirements (for Systems and Equipments)."
MIL-STD-471 - Maintainability Demonstration.
MIL-STD-785 - Requirements for Reliability Program (for Systems and Equipments).
MIL-STD-1399, Section 103 - Electric Power, Alternating Current.

"HANDBOOKS

MILITARY
MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment.
MIL-HDBK-472 - Maintainability Demonstration.

"PUBLICATIONS

MILITARY
0967-312-8010 - Maintainability Design Criteria Handbook for Design of Shipboard Electronic Equipment.
0367-048-0010 - Maintainability Design Criteria Handbook for Designers of Shipboard Electronic Equipment.
0969-019-7000 - Electronic Test Equipment Application Guide."

Pages 6 and 7

3.2.23, item (a), line 1: Delete "any number (up to 4) of" and substitute "two or three".

3.2.23.1: In line 7, delete the colon after "sources" and substitute a period; delete items (a) and (b).

1/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

3.2.5: Delete "Load rating (3.2.1(b)(1))".

3.3.2 through 3.3.4: Delete and substitute:

"3.3.2 Reliability. To meet the needs of the Naval service, it is imperative that reliability of operation be considered of prime importance in the design and manufacture of the equipment. The supplier shall employ all reasonable methods possible in the process of manufacture to assure quality and maximum reliability. In order to insure high equipment reliability under all service conditions to avoid marginal applications and provide adequate safety factors, it may be necessary to derate certain parts from their rated values when applying them to equipment circuits. The design shall include all possible features which will result in reliable and stable operation with minimum need for adjustment and alignment, and with Mean-Time-Between-Failures (MTBF) not less than 2500 hours for the entire motor generator and associated control system.

"3.3.2.1 Equipment failure is defined as occurring when either the equipment ceases to function or its performance degrades below the requirements of this specification.

"3.3.2.2 The MTBF shall be predicted in accordance with MIL-HDBK-217. The section "How to Make a Reliability Prediction Based on Stress Factors and Part Populations" shall be used for the method of prediction and report format. The most critical items shall be identified in reliability prediction in accordance with MIL-STD-785 (see 6.3).

"3.3.2.3 Failure rate data used shall be as given primarily in MIL-HDBK-217. Failure rate data from other sources or estimated failure rates, if necessary, shall be submitted to the procuring activity for review prior to use. In the event other sources than quoted herein are used for failure rates derivation, detailed substantiation shall be given. No acceleration factors shall be used unless justified by special quality assurance controls and justification submitted.

"3.3.2.4 Stress levels shall be calculated from design data in accordance with MIL-HDBK-217. The operating electrical and thermal stress levels shall be obtained during the tests of 4.6.21 and 4.6.22. The stress levels obtained by test shall be compared to the application ratings (maximum stress levels allowed) given by part or unit vendors and the calculated stresse levels using tabular form. Where parts are found by test or calculation to be stressed beyond the application rating (maximum stress level) necessary design changes shall be made and stress level tests shall be repeated. Where the test and calculated values differ by more than 10 percent a second stress calculation shall be made on that part to determine cause of discrepancy.

"3.3.2.5 The supplier shall submit the Reliability Predictions and definition of failure in terms of actual equipment configuration to the procuring activity for review.

"3.3.3 Maintainability. The supplier shall establish the maintainability program which shall be integrated with the equipment design engineering program. The maintainability program shall be consistent with the type and complexity of equipment, and shall contain but shall not be limited to the following:

(a) Maintainability program plan.
(b) Maintainability analysis and corrective action system.

"3.3.3.1 The maintainability shall be defined in terms of Geometric-Mean-Time-To-Repair ($MTTR_G$). $MTTR_G$ for the equipment shall be 2 hours maximum. Equipment repair time (ERT) for any one item shall not exceed 6 hours.

"3.3.3.2 The supplier shall define in the Maintenance Program Plan how he will develop and conduct the maintainability program to meet the quantitative requirements. He shall compute $MTTR_G$ predictions and submit them to the procuring activity for review.

"3.3.3.3 The $MTTR_G$ shall be predicted in accordance with MIL-HDBK-472, maintainability prediction procedure II. The failure rates used shall be the same as used for the reliability prediction. Wherever the expected life of the part or unit is shorter than the expected life of the equipment, the replacement of the part or unit shall be scheduled in the preventive maintenance section of the technical manual for the equipment so as to prevent equipment failures due to wear out of its parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

Pages 6 and 7 (cont'd.)

"3.3.3.4 Preventive maintenance shall be simplified so as to reduce the need for highly skilled maintenance personnel. The complexity of maintenance shall be reduced by providing adequate accessibility, and clearance, and by providing for interchangeability of like units and parts. All essential parts shall be interchangeable physically and electrically with corresponding repair parts without selective matching, drilling, filing, or use of undue force.

"3.3.3.5 Corrective maintenance downtime shall be reduced by designing for rapid and positive detection of malfunction or degradation (implementation of indicators, test points, and so forth), ease of fault corrections, and rapid positive verification of correction.

"3.3.3.6 Equipment shall be subjected to a maintainability demonstration test. Sample size of corrective maintenance actions shall be 20, sample selection in accordance with the "Task Selection" appendix, "Corrective maintenance sample selection" table of MIL-STD-471 (see 6.3). $MTTR_G$ shall be computed by the following formula:

$$\log MTTR_G = \frac{\sum_{i=1}^{20} (\log M_{ci})}{20}$$

where M_{ci} is the repair time of corrective maintenance action "i" repair time includes fault location, isolation, replacement of part(s) and realignment or readjustment as necessary through final checkout.

"3.3.3.6.1 A maintenance test plan shall be prepared and submitted to the procuring activity for review before the tests commence. The plan shall define in detail the maintainability actions to be performed during the test. The plan shall provide for random selection of failure events to be chosen at the time of the test.

"3.3.3.6.2 A report of the maintainability test shall be prepared by the supplier and submitted to the procuring activity.

"3.3.3.7 Data collection. The supplier shall establish and maintain a data collection system covering failure modes and effects, analysis, and corrective action throughout all stages of development, production and testing of the equipment. Any failure that occurs shall be thoroughly analysed and recorded in the log book. The corrective action shall be implemented and design review shall be initiated as necessary to improve the reliability and maintainability of the equipment and to meet the quantitative requirements. All failures and actions taken shall be reported to the procuring activity in two reports: (a) prior to first delivery and (b) with the last equipment delivered.

"3.3.3.8 The control equipment shall be designed to permit easy installations and to keep maintenance requirements to a minimum, both in level of skills and in number of personnel required to maintain the equipment. The equipment shall also be designed so that it can be maintained by replacement of easily removed parts, modules, or subassemblies. The design shall follow design for maintainability considerations in MIL-STD-470 (see 6.3) and NAVSHIPS 0967-312-8010.

"3.3.3.9 Human engineering requirements. Human engineering requirements shall be in accordance with MIL-H-46855 (see 6.3).

"3.3.3.10 Test equipment. The motor generator shall be so designed as to be capable of being maintained, aligned, calibrated, and repaired with the aid of the following test equipment:

(a) Clamp-on volt-ammeter - 0 to 600 amp and 0 to 600 volts a. c. range, FSN: 1N6625-284-0855 or 1N6625-879-1231, MTN: CV-633 or AN/USM-33.
(b) Multimeter - 0 to 4000 volts d. c., 0 to 800 volts a. c., 0 to 16 amps d. c. and 0 to 200 megohm range; FSN: 1N6625-684-3082; MTN: AN/USM-123; G. E. Simpson Model 269 or equivalent.
(c) Oscilloscope - FSN: F6625-787-0309, MTN: AN/USM-117.
(d) D. c. milliammeter - 0 to 150 ma range; FSN: 2N6625-945-2759; MTN: CV 931-4904-003.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

(e) A. c. voltmeter - 0 to 600 volts range, FSN 1N6625-184-5337; MTN: MR71W600ADVV in accordance with MIL-M-16034.
(f) An isolated variable voltage d. c. power supply 0 to 35 volts (0 to 3 amp capacity).

Suppliers desiring to utilize test equipment other than specified herein, shall submit to the procuring activity with justification, the necessity for using such test equipment, prior to the start of manufacture of the equipment. Test equipment required for maintenance not listed herein, in the Portable Test Equipment list for Electric Plant, or in NAVSHIPS 0969-019-7000 shall be furnished by the supplier with the motor generator.

"3.3.3.11 Plan for use. Unless otherwise specified (see 6.1), reliability and maintainability calculations shall be based on average operating time of 4 days per week or 139 days per year. System characteristics shall be as specified for type II power of MIL-STD-1399, Section 103. Loads shall be considered to vary between one-quarter, one-half and full load steps changing at 1 hour intervals."

3.3.5.2, line 7: Delete "type IC/N31A of"; line 8, after "more": Add new sentence: "Thermostatic switch shall be adjustable between 90° and 165°F. and shall be capable of withstanding 195°F. without damage."

Page 11

3.4.6.1: Delete and substitute:

"3.4.6.1 Radial, single row, both races extended, sealed, cartridge type, type 120 in accordance with FF-B-171 shall be used up to and including size 315. Sizes 316 and larger shall be single shielded, type 111, class 2. When cartridge type 120 bearings are used, a plate bearing the warning 'DO NOT LUBRICATE' shall be secured to the end brackets."

Page 17

Add as new paragraph 3.5.2.5:

"3.5.2.5 Test points. Test points in accordance with MIL-STD-415 shall be provided for items (b) and (c) of 3.5.2.1. Sufficient test points shall be brought out to a jack board in order to monitor equipment operation while performing all facets of equipment trouble shooting. These test points shall also be capable of being utilized during normal operation without adversely affecting system performance. Test points for voltage readouts shall not be utilized if a voltmeter is installed for continuous readings. The test points shall be readily accessible for use by service personnel. Each circuit point shall be clearly identified. In addition to individual jacks for each point, the points may also be connected through a selector switch or switches to a single jack."

Page 18

3.5.3.11, last sentence: Delete and substitute "The control equipment shall be designed to permit easy installations and to keep maintenance requirements to a minimum, both in level of skills and in number of personnel required to maintain the equipment. The equipment shall also be designed so that it can be maintained by replacement of easily removed parts, modules, or subassemblies. The design shall follow design-for-maintainability considerations in MIL-STD-470 (see 6.3) and NAVSHIPS 0367-048-0010."

Pages 19 and 20

3.5.4.8, item (b), second sentence: Delete and substitute "A relay may be used in the excitation circuit to provide generator field flashing upon starting only."; item (c) (1) a, line 1: Delete "energized automatically" and substitute "provided and connected to automatically operate"; after first sentance, add: "Relay shall be suitable for connection so that when generators are not being operated in parallel, no electrical connections will exist between generators or associated controls."

Page 21

3.5.4.11, delete and substitute:

"3.5.4.11 Circuit breakers. Circuit breakers, when required, (see 3.5.2.1) shall be type ABB conforming to MIL-C-17361. Circuit breakers shall be provided with a "G" element (generator protection) and shall include an auxiliary switch and a 115 volt, d. c. shunt trip device."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

Page 21 (cont'd.)

3.5.4.15, add new sentence: "An off-on switch shall be provided to disconnect the synchroscope from the line when not in use."

Add as new paragraph 3.5.4.18:

"3.5.4.18 Reverse power protection. The shunt trip on the circuit breaker shall be electrically interlocked with the motor controller in such a manner that deenergizing the motor controller will result in the activiation of the shunt trip on the circuit breaker."

Page 24

3.7.3, item (d): Delete "(5 sheets maximum)."

3.7.3, add as item (g):

"(g) Water cooler drawing assembly and details (2 sheets maximum)."

Pages 26, 27, 28 and 29

3.7.3.4: In item (c) (1)c, after "terminals" insert "and test points"; and in item (d) (1)b, after "subassemblies" insert "and all test points".

Pages 30 and 31

3.8, item (g) (3): In line 3, delete "Open loop test" and substitute "Open loop tests-Give detail instruction on performing open loop tests and explain how the results of these tests are used in localizing faults in the equipment"; and delete lines 5 and 6 and substitute "Tables showing test points of measurements of voltage, current, resistance and a brief description of how and when each test point should be used."

4.1, delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

4.2, last line: Add "(see 6.1 and 6.3)".

4.3.2, line 8: Delete "inspector" and substitute "Government".

Page 33

4.4.1, first sentence: Delete and subsitute:

"4.4.1 Test results shall be recorded by the supplier on test forms prepared by the supplier."

4.4.3, delete and substitute:

"4.4.3 All test data shall be witnessed and verified by a qualified test engineer prior to the shipment of the equipment. Upon completion, one file copy of all periodic test data shall be forwarded to the command or agency concerned."

Page 34

4.6.7.2, line 3: Delete "inspector" and substitute "Government".

Pages 40 and 41

4.6.22.2, line 9: Delete "table I" and substitute "table II".

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

Pages 40 and 41 (cont'd.)

4.6.22.4, item (h): Delete "for all parts".

4.6.22.5: Add "Heating tests shall be made only with the equipment completely assembled and mounted in the manner for which it is designed. Barriers shall be placed adjacent to the enclosure, 2 inches from the back and each side. Heating test readings shall be taken on all coil windings (including power transformers and reactors) and rectifiers except for those rectifiers of control circuits which clearly do not contribute to the overall temperature rise within the equipment enclosure and except for those which are tested and given separate qualification approval under its individual specification. For those parts which are given exception based on separate qualification approval, it shall be determined that its adjacent ambient within the enclosure does not exceed the rated ambient of the part and that the application rating does not exceed the manufacturer's approved rating at the applied temperature ambient."

Pages 45/46

6.1, add as items (h) and (i):

"(h) Reliability and maintainability calculation required (see 3.3.3.11)."
"(i) Quality assurance requirements (see 4.2).

Add as 6.3:

"6.3 Management control system documents. The following management control system documents should be included on DD Form 1660:

(a) MIL-STD-785 (see 3.3.2.2).
(b) MIL-STD-471 (see 3.3.3.6).
(c) MIL-STD-470 (see 3.3.3.8 and 3.5.3.11).
(d) MIL-H-46855 (see 3.3.3.9).
(e) MIL-Q-9858 (see 4.2)."

Page 47

10.1, last sentence: Delete and substitute:

"This list is not an all inclusive list and additions and deletions as applicable can be made by the manufacturer."

30.1.2, lines 2 and 3: Delete "Government".

Preparing activity:
Navy - SH
(Project 6125-N152)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.